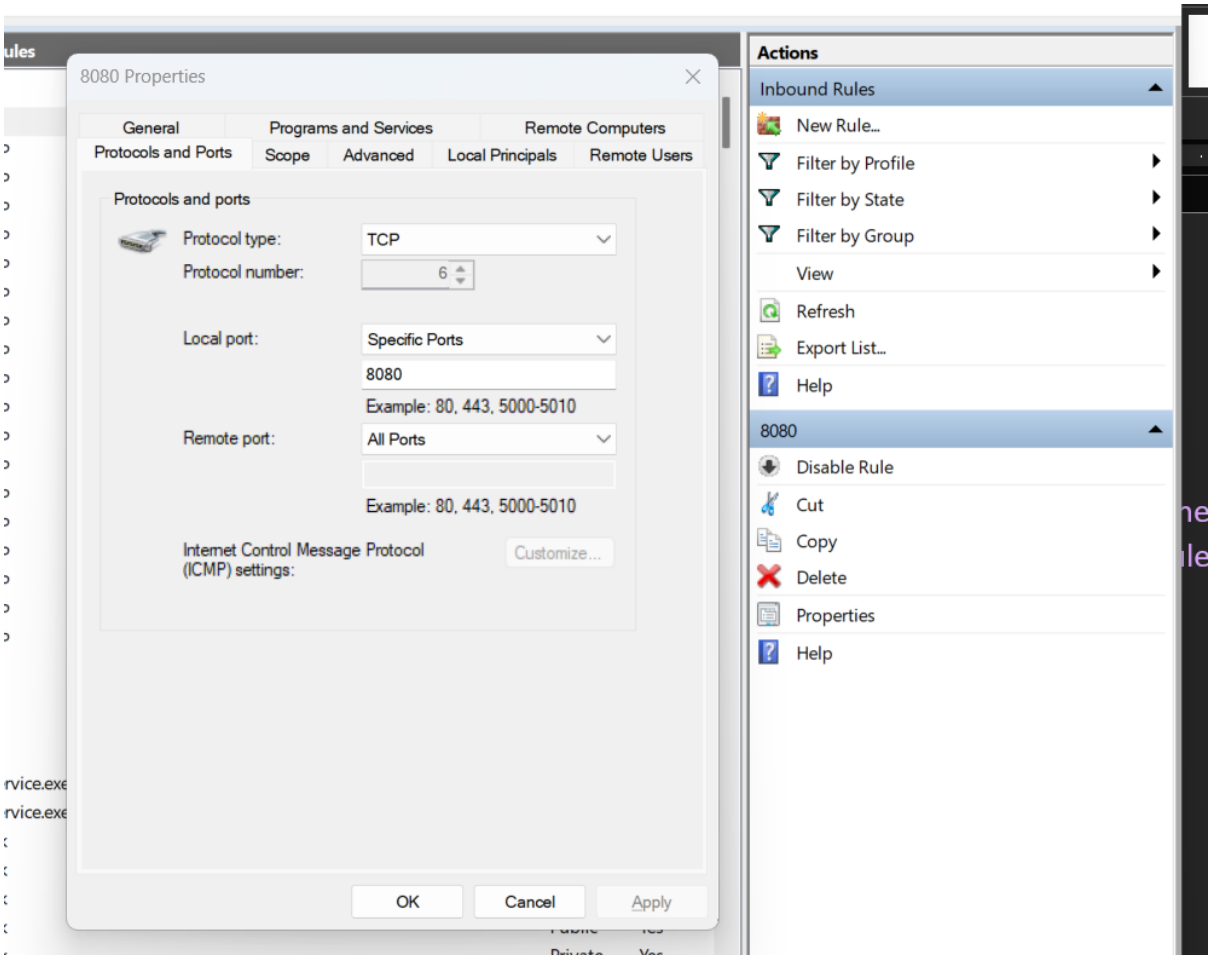
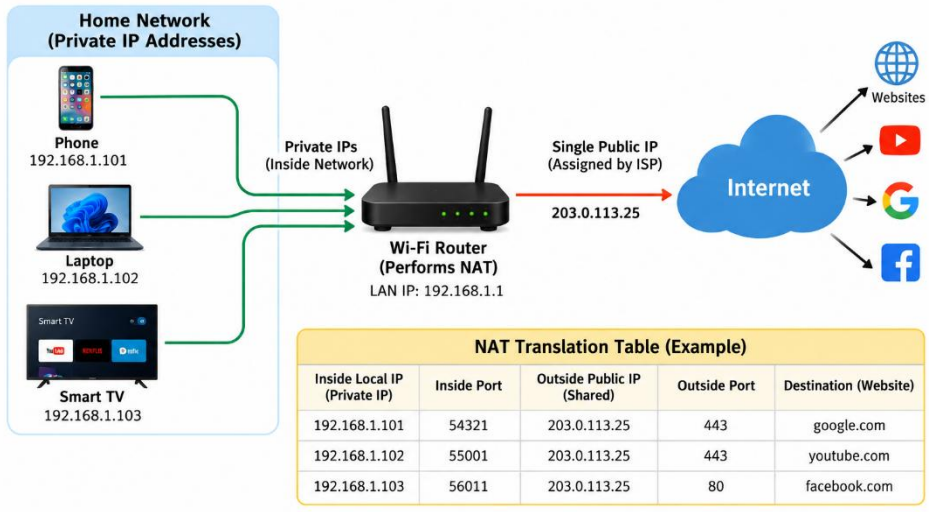
secure networks with firewalls

1. Set up Windows Defender Firewall on your Windows laptop or PC and create a new inbound rule to block all incoming traffic on port 8080. Take a screenshot of the rule you created.



2Simulate NAT (Network Address Translation) by drawing a simple diagram showing how your home WiFi router assigns private IP addresses to your phone, laptop, and smart TV, and how it translates them to a single public IP for internet access.

NAT (Network Address Translation) Simulation



NAT Translation Table (Example)

Inside Local IP (Private IP)	Inside Port	Outside Public IP (Shared)	Outside Port	Destination (Website)
192.168.1.101	54321	203.0.113.25	443	google.com
192.168.1.102	55001	203.0.113.25	443	youtube.com
192.168.1.103	56011	203.0.113.25	80	facebook.com

When devices in the **home network** (192.168.1.x) access the internet, the router (NAT) replaces their private IP addresses with its single public IP (203.0.113.25).
The router keeps a translation table to send the responses back to the correct device.

IP Address Table

Device	IP Address	Type
Phone	192.168.1.101	Private IP
Laptop	192.168.1.102	Private IP
Smart TV	192.168.1.103	Private IP
Wi-Fi Router (LAN)	192.168.1.1	Private IP
Wi-Fi Router (Internet)	203.0.113.25	Public IP (example)

Working

1. The Wi-Fi router assigns a unique private IP address to each device using DHCP.
2. When a device accesses the internet, NAT translates its private IP address to the router's public IP address.
3. All three devices share the same public IP address to access websites.
4. The router tracks connections using port information and forwards returning data to the correct device.

Conclusion: NAT allows multiple devices on a home network to access the internet through a single public IP address while keeping their private IP addresses within the local network.

3. Configure port filtering using your WiFi router's admin panel (or use a router simulator online if you don't have access). Block outgoing traffic on port 25 (SMTP) and explain in 2 sentences how this could help prevent spam from your network.

Port Filtering Configuration – Blocking Port 25 (SMTP)

Objective: Configure the Wi-Fi router to block outgoing traffic on port 25 to help prevent spam emails from being sent through the network.

Steps to Configure

1. Connect your laptop to your Wi-Fi network.
2. Open a browser and enter your router's gateway IP, usually 192.168.1.1 or 192.168.0.1.
3. Log in using your router's admin credentials.
4. Navigate to Firewall, Security, or Access Control → Port Filtering (the menu varies by router).
5. Add a new filtering rule with these settings:

Setting	Value

Rule Name	Block SMTP
Direction	Outgoing / Egress
Protocol	TCP
Source IP	All devices / Any
Destination Port	25
Action	Deny / Block
Status	Enabled

6. Save or apply the settings and verify that the rule is enabled.

How This Helps Prevent Spam

Blocking outgoing TCP port 25 prevents devices on the network from directly sending email through standard SMTP connections, reducing the risk of infected devices sending spam emails. However, it does not block all spam because attackers may use alternative ports, such as 587 or 465, or external email services.

Result: Outgoing SMTP traffic on port 25 is blocked by the router's filtering rule.

4.Given a sample packet filtering rule set (provided by your trainer), identify which of these three sample packets would be allowed or blocked: 1) HTTP request from 192.168.1.5 to 172.217.166.78:80, 2) SSH request from 10.0.0.10 to 34.201.23.1:22, 3) DNS query from 192.168.1.7 to 8.8.8.8:53.

Packet Filtering: Allowed or Blocked

Objective: To identify whether packets are allowed or blocked based on source IP addresses, destination IP addresses, and port numbers.

Since the trainer's packet filtering rule set was not provided, the following is an example using common firewall rules.

Sample Packet Analysis

No.	Source IP	Destination IP	Port	Protocol	Decision
1	192.168.1.5	172.217.166.78	80	HTTP (TCP)	Allowed
2	10.0.0.10	34.201.23.1	22	SSH (TCP)	Blocked
3	192.168.1.7	8.8.8.8	53	DNS (UDP)	Allowed

Explanation

1. HTTP (Port 80): Allowed — HTTP traffic is permitted by the example firewall rules.
2. SSH (Port 22): Blocked — SSH traffic is denied by the example firewall rules.

3. DNS (Port 53): Allowed — DNS queries using UDP port 53 are permitted by the example firewall rules.

Conclusion: Packet filtering checks IP addresses, protocols, and port numbers to decide whether network traffic should be allowed or blocked. These decisions are illustrative; use your trainer's actual rule set for the correct answers.

5.Research and list 3 ways Instagram or Zomato might use firewalls and packet filtering to protect their servers from attacks or misuse.

3 Ways Instagram or Zomato Use Firewalls and Packet Filtering

1. Blocking Unauthorized Access: Instagram and Zomato can use firewalls to allow traffic only on required ports, such as HTTPS (port 443), and block unauthorized connections to internal servers.

2. Preventing Cyberattacks: Firewalls and packet filtering can block traffic from suspicious IP addresses and restrict unwanted ports or protocols, helping reduce the risk of attacks such as port scanning and unauthorized access.

3. Protecting Against Misuse and Traffic Floods: These platforms can combine firewalls with rate limiting and DDoS protection to detect or restrict excessive requests that may overload their servers or disrupt services.

Conclusion: Firewalls and packet filtering help protect Instagram and Zomato by controlling network traffic, blocking suspicious connections, and improving server security. These are common security practices; the exact configurations used by each company are not necessarily public.